

Appendix A

Industry Internship Report

On

Developing a web portal for management & viewing of manuals.

Submitted to

Amity University Uttar Pradesh



in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

(CSE)

By

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under the guidance of

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Abstract

The NHPC Manuals Directory is a centralized, web-based platform developed to enhance the accessibility and management of technical manuals across various departments within NHPC Ltd. Traditionally, retrieving manuals involved navigating through unstructured folders or relying on physical copies, which often led to delays and inefficiencies. This project addresses these challenges by introducing a structured and user-friendly digital interface where users can browse categorized manuals, perform keyword searches, and download relevant documents in PDF format.

The system is built using a full-stack development approach. The frontend, developed with HTML, CSS, JavaScript, and Bootstrap, ensures a responsive and intuitive user experience. The backend, powered by Node.js and Express, handles API requests and communicates with a MySQL database that stores metadata for each category and manual. Manuals are securely stored in a static folder and served directly upon user request. Features like search filtering, organized category views, and clean navigation further improve usability.

This project not only simplifies documentation access but also lays the groundwork for scalable document management in the future. By digitizing manual storage, it significantly reduces time spent locating technical documents, supports efficient operations, and contributes to the broader goal of digital transformation within NHPC.

Table of Content

S.NO	TOPIC
1	Duration
2	Place
3	Introduction
4	Profile of Organization
5	Project Descriptions
6	Activities Undertaken
7	Tools and Techniques Applied
8	Observation
9	Evaluation
10	Learning
11	Reflection
12	Conclusion

I. Duration / No. of Days: 15 May 2025 – 04 July 2025 (7 Weeks)

II. Place: NHPC Office Complex, Santosh Nagar, Sector 33, Faridabad, Haryana 121003

III. Introduction

As part of the Bachelor's curriculum in Computer Science engineering (ASET) in Amity University Noida, a two-month industry internship was mandated after the end of 6th semester to integrate classroom learning with practical experience. This report encapsulates my 7-week internship at NHPC Limited, where I developed a web portal for management & viewing of manuals.

The project catered to my interest in frontend & backend (web development). I was tasked with building a web portal for management & viewing manuals.

Learning Goals:

- Acquire insight into NHPC's IT systems and operational processes.
- Develop hands-on experience in a professional software development setting.
- Translate academic concepts into practical solutions for data protection issues.
- Strengthen abilities in task scheduling, concurrency handling, and database operations.
- Build communication and teamwork skills within a technical environment.

This report presents the organization's background, a summary of the project, a description of the tasks carried out, key observations, and personal reflections, concluding with an assessment based on the defined learning objective.

Objective

To design a web-based directory system that allows users to efficiently browse, search, and download categorized NHPC manuals in PDF format, ensuring quick access to critical technical documents.

Project Overview

The NHPC Manuals Directory is a user-friendly platform developed to organize and display various technical manuals under predefined categories such as Electrical, Mechanical, Civil, and more. The system enables users to search for categories, view manuals within each category, and directly download them for offline use

Background

NHPC Ltd., being a leader in hydropower development, relies on extensive documentation for operations, safety, maintenance, and compliance. Traditionally, accessing these manuals required physical copies or scattered digital folders, leading to inefficiencies and accessibility issues

Motivation

The primary motivation behind this project is to simplify document access for NHPC engineers, staff, and stakeholders by creating a centralized, searchable, and responsive web interface. This minimizes time spent locating files and supports better information management.

IV. Profile of Organization

Name of Organization: National Hydroelectric Power Corporation (NHPC) Limited

Address: NHPC Office Complex, Santosh Nagar, Sector 33, Faridabad, Haryana 121003

Mission and Objectives:

- Plan and promote efficient development of hydropower in India.
- Expand into solar and wind to foster sustainable growth.
- Adopt digital transformation for operational excellence and data security.

V. Project Descriptions

Project: NHPC MANUALS DIRECTORY.

The **NHPC Manuals Directory** project aimed to develop a responsive and user-friendly web application to manage and access NHPC's technical and operational manuals efficiently. The

primary goal was to create a centralized digital platform where users could browse categorized manuals and admins can manage the documents and manuals.

The frontend was developed using **HTML**, **CSS**, and **JavaScript** to structure and style the content. **Bootstrap CDN** was integrated to ensure a responsive design, consistent styling, and improved layout across different devices. For enhanced interactivity and dynamic DOM manipulation, **jQuery** was utilized.

The backend was built using **Node.js** and **Express.js**, providing a lightweight and efficient server-side environment. The backend handled HTTP requests through **POSTMAN**, served static files, and communicated with the database to fetch and manage manual data. A **MongoDB** or **MySQL** database (depending on configuration) was used to store manual categories, file paths, and metadata.

Methodology

The project was built using a full-stack web development approach:

- **Frontend:** HTML, CSS, and Bootstrap for responsive design and UI components.
- **Backend:** Node.js and Express for serving category and manual data.
- **Database:** MySQL for storing category and manual metadata.
- Manual files are stored as downloadable PDFs, and category filtering is enabled via dynamic search and click-based routing.

Files and Folders Description

The project is organized into the following files and directories to ensure modularity and maintainability:

Frontend Folder (nhpc-mannuals/frontend)

contains all files related to the user interface.

- **index.html** – (Optional landing page) Entry point or welcome page.
- **categories.html** – Displays all manual categories in a card layout with a search bar.
- **categories_files.html** – Displays all manual files for a selected category with clickable download links.

- **style.css** – Custom styling for buttons, layout, background gradients, and responsiveness.
- **script.js** – Handles dynamic behavior like fetching categories/files from backend and rendering them.

Backend Folder (nhpc-manuals/backend)

Handles data serving and API interactions.

- **server.js** – Main server file using Node.js + Express. It connects to the MySQL database and serves API endpoints for categories and manual files.
- **routes/**
 - **manualRoutes.js** – Defines API routes for fetching categories and their corresponding files.
- **controllers/**
 - **manualController.js** – Handles logic for retrieving data from the database and sending responses.
- **public/** – Serves static files like manuals (PDFs), images, and CSS/JS for frontend access.

Database Folder

- **nhpc.sql** – SQL dump file that contains the schema and initial data for categories and manuals tables.

Manuals Folder (backend/public/manuals)

Stores all downloadable PDF manuals categorized by folder or flat list.

Example files:

- turbine1.pdf
- compressor.pdf
- valve.pdf

VI. Activities Undertaken

- Collected requirements from NHPC IT staff regarding manual categorization, search functionality, and accessibility.
- Designed the overall architecture using HTML, CSS, JavaScript, jQuery, Bootstrap (via CDN), and a Node.js + Express backend.
- Implemented dynamic API routes in Node.js to serve category-specific manuals based on user interaction.
- Utilized jQuery and AJAX to fetch and render manuals in real-time without reloading the page.
- Built a RESTful API for manual retrieval and category-level filtering.

VII. Tools and Techniques Applied

- **HTML,CSS,JavaScript,Bootstrap(CDN):**
Used to build the frontend interface of the NHPC Manuals Directory. HTML structured the content, CSS handled styling, and JavaScript enabled interactivity. Bootstrap via CDN was utilized for responsive design and consistent UI components across devices.
- **Node.jsandExpress.js:**
Formed the core backend framework for handling HTTP requests, serving APIs, and managing routing logic. Express.js enabled efficient development of RESTful API endpoints for fetching and displaying manuals by category.
- **jQueryandAJAX:**
jQuery simplified DOM manipulation and event handling, while AJAX was used to make real-time API calls without reloading the page. This allowed a seamless user experience when browsing and viewing manual categories.
- **JSON-basedDataHandling:**
Manual information was structured using JSON, allowing dynamic updates and retrieval of manuals through the backend with minimal overhead. This format ensured easy parsing and integration with frontend scripts.

- **Git(VersionControl):**

Git was used to manage and track changes in the project's codebase. It enabled version control, collaboration, and rollback functionality during development and testing phases.

- **CommunicationandCollaboration:**

Actively engaged with teammates and stakeholders to gather functional requirements, discuss design ideas, and present progress updates. Clear communication ensured the project stayed aligned with NHPC's internal needs and usability expectations.

VIII. Interface Previews and System Snapshots

A.) NHPC MANUALS DIRECTORY

1. MANUALS CATEGORIES

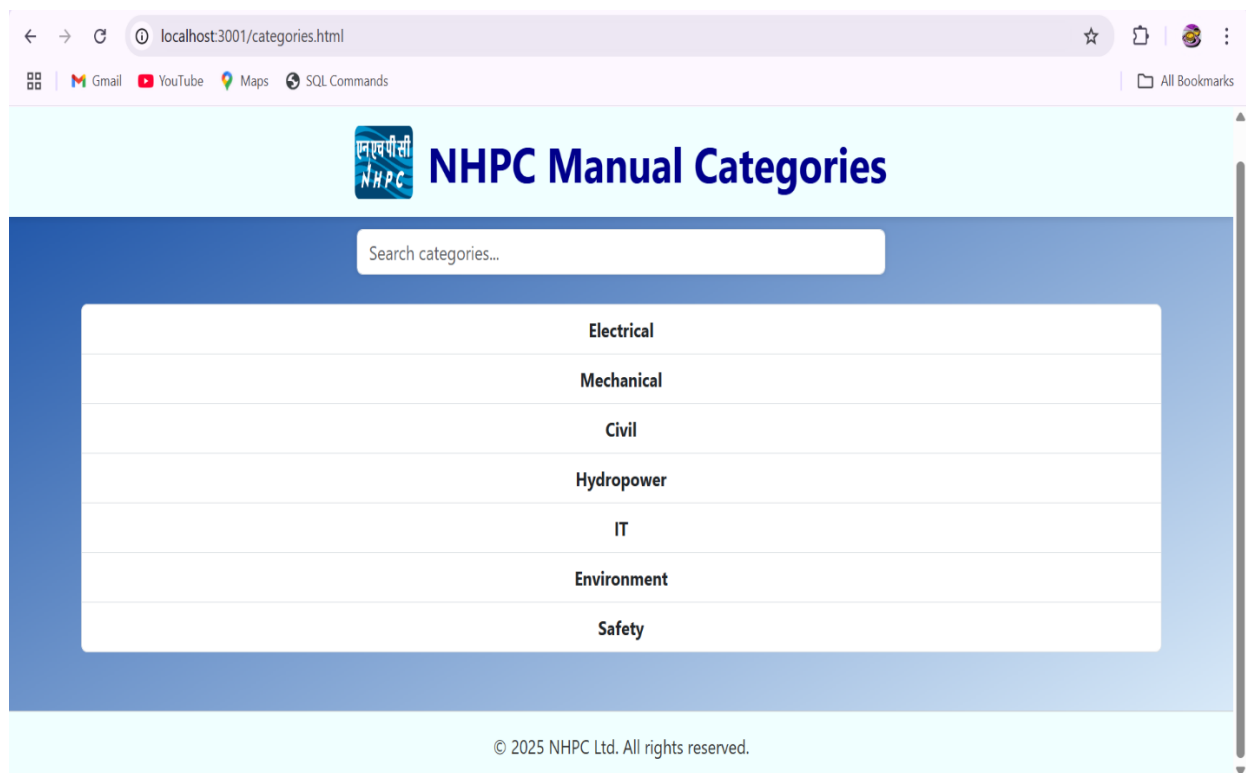


Figure1: NHPC Manual Categories Web Interface

The above interface represents the main category selection page of the NHPC Manuals Directory system. Users are presented with a clean, user-friendly dashboard that allows them to browse

manuals by various categories such as Electrical, Mechanical, Civil, Hydropower, IT, Environment, and Safety. The page features a dynamic search bar to quickly filter categories and is styled with a blue gradient background that aligns with NHPC's official branding. Clicking on any category navigates the user to a new page where they can access and download manuals relevant to that category in PDF format. This structure ensures efficient navigation and easy access to technical documentation for NHPC personnel and stakeholders.

2. CATEGORIES FILES

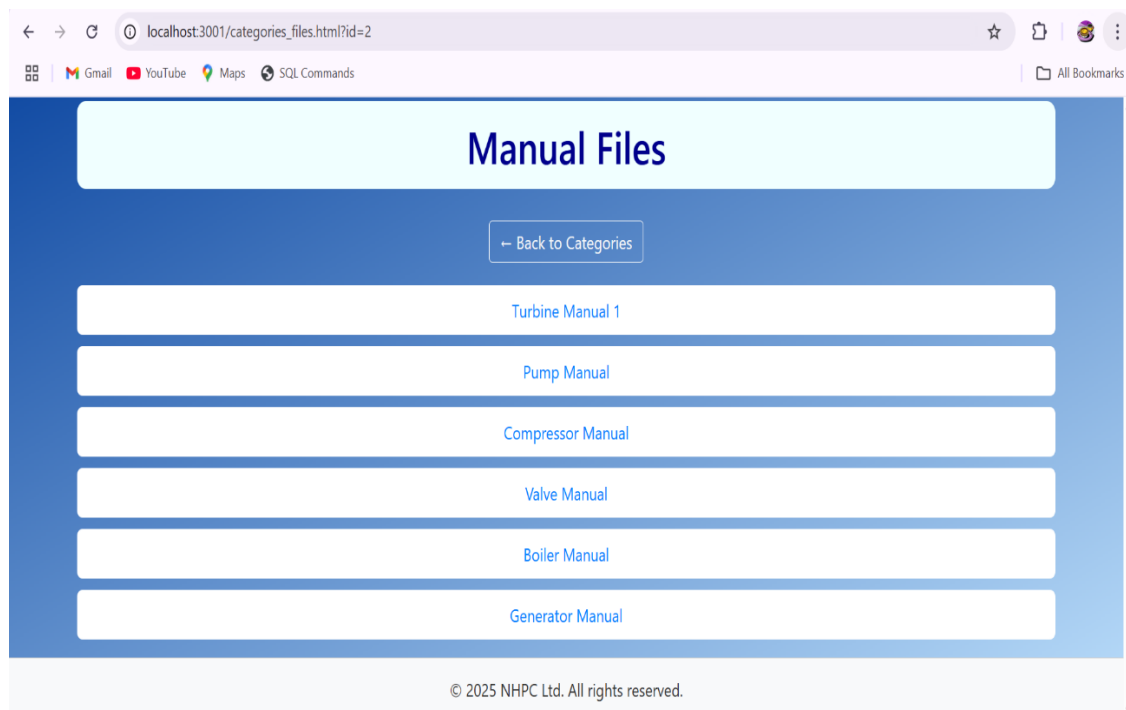


Figure 2: Manual Files Listing for Selected Category

This page displays all available manuals under a selected category from the NHPC Manuals Directory. Once a user clicks on a category from the previous page, they are redirected here to view specific manual files such as the *Turbine Manual*, *Pump Manual*, *Compressor Manual*, and others. Each manual name functions as a clickable link, enabling users to easily download the corresponding PDF file. A "Back to Categories" button is provided for smooth navigation, ensuring users can return to the main category page without refreshing or restarting their session.

The consistent design theme and structured layout improve usability and enhance the user experience.

3. FILES DOWNLOAD

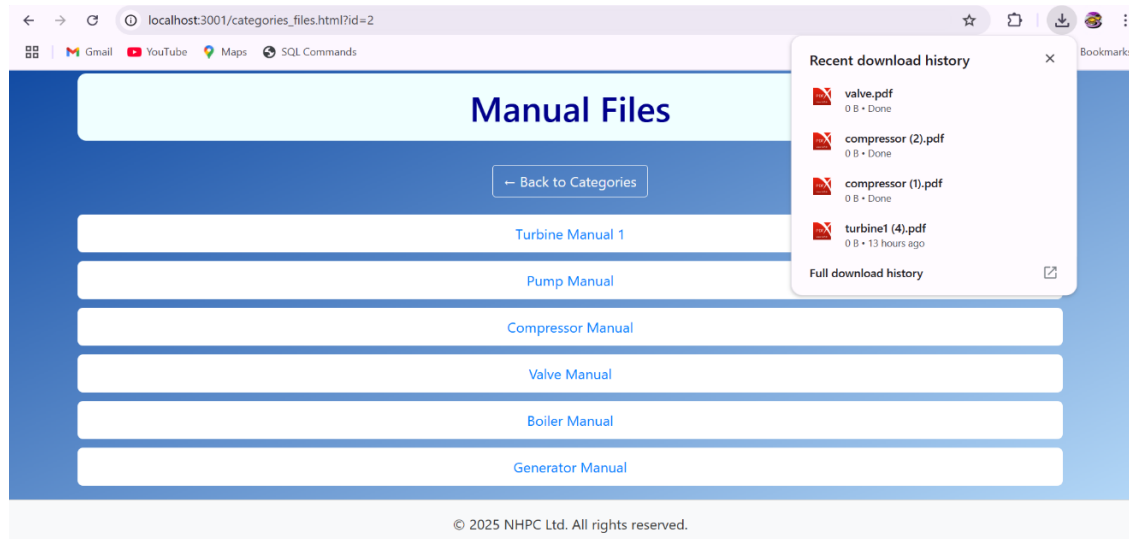


Figure 3: Manual Download Confirmation and Functionality

This screenshot highlights the successful download capability of the NHPC Manuals Directory. After clicking on any manual link, such as *Valve Manual* or *Turbine Manual*, the corresponding PDF file is instantly downloaded to the user's system, as seen in the browser's recent download history. This feature ensures that critical documents are readily accessible offline, supporting operational efficiency. Each manual is neatly organized and named for easy identification. The design emphasizes quick access, seamless interaction, and an intuitive experience for NHPC personnel managing or referencing technical manuals.

VIII. Observation

- **Categorical organization enhances clarity and user navigation:** Displaying all available categories upfront allowed users to quickly identify and access specific sections such as Mechanical, Electrical, or Civil manuals, making the interface intuitive and structured.
- **Category-based routing improves content management:** Clicking on a category dynamically loads a new web page that displays only the manuals

related to that category. This separation keeps the interface clean and prevents information overload.

- **Manual download functionality increases system utility:**
Users can click on any listed manual to instantly download it in PDF format. This feature made the system highly practical for field engineers and internal teams needing quick offline access.
- **Clean UI built with Bootstrap encourages adoption by non-technical users:**
Thanks to responsive design and a visually clean layout, even users with minimal technical experience found the system easy to use.
- **Centralization reduces dependency on physical documents or scattered files:**
Having all manuals stored and accessible through a single platform eliminates the hassle of tracking down physical copies or digging through folders.
- **Scalable architecture enables future enhancements:**
The system is flexible enough to integrate additional features later, such as admin uploads, search filters, user authentication, or cloud-based storage support.
- **Security and access control considerations are critical for sensitive documents:**
Though not yet fully implemented, the need for authentication and role-based access was identified to protect confidential manuals from unauthorized downloads.

IX. Evaluation

- **Successfully met core project objectives:**
The NHPC Manuals Directory was designed and implemented as a centralized platform to access categorized manuals, enabling easy navigation, real-time loading, and PDF downloads—all of which aligned with the project's initial goals.
- **Practical application of full-stack web development skills:**
The project allowed hands-on experience with frontend (HTML, CSS, JavaScript, jQuery, Bootstrap) and backend (Node.js, Express.js) technologies, improving both technical understanding and implementation ability in a real-world context.
- **Improved understanding of REST API design and dynamic data handling:**
Creating and consuming custom API endpoints to fetch manuals by category and enable dynamic rendering strengthened knowledge of asynchronous programming and client-server communication.
- **Learned the importance of maintainable and scalable architecture:**
The modular design of the system makes it easy to extend in the future—such as adding admin upload panels, advanced search filters, or authentication features—demonstrating good development foresight.
- **Developed professional communication and documentation skills:**
Regular discussions with peers and mentors helped refine requirements, while deployment instructions and user guidelines ensured ease of use and future maintenance.
- **Identified future scope and security concerns:**
The evaluation revealed the need for enhanced access control and secure download permissions, especially in environments dealing with sensitive documentation like NHPC.

XI. Reflection

This project has been a truly enriching experience, bridging the gap between academic knowledge and real-world application. Developing the NHPC Manuals Directory from the ground up challenged me to think beyond just writing code—it required a deep focus on user accessibility, interface responsiveness, and system scalability. Designing an intuitive platform that categorizes manuals and supports real-time PDF downloads taught me the importance of structuring information clearly for diverse users, including non-technical staff.

Working with technologies like Node.js, Express, jQuery, and Bootstrap improved my understanding of full-stack development and real-time client-server interactions. Facing challenges such as dynamic data loading, UI responsiveness, and future-proof architecture strengthened my problem-solving and debugging skills. This experience also emphasized the significance of user-centered design, version control, and clean code practices.

Most importantly, the project reinforced the value of collaboration, clear communication, and adaptability in a professional environment. It confirmed my interest in web development and system integration and gave me the confidence to take on more complex, real-world software challenges in the future.

XII. Conclusion

The NHPC Manuals Directory project successfully delivered a centralized, user-friendly platform for accessing and downloading categorized technical manuals. By integrating modern web technologies like HTML, CSS, JavaScript, Bootstrap, jQuery, and Node.js, the system achieved its goal of improving document accessibility, streamlining manual retrieval, and enhancing the overall efficiency of NHPC's internal operations.

The project not only met functional requirements—such as category-based navigation, real-time content loading, and PDF download support—but also laid a strong foundation for future enhancements like admin uploads, authentication, and advanced search capabilities. This experience provided valuable insights into full-stack web development, RESTful API integration, and responsive UI design.

Overall, the project proved to be a practical, impactful solution for document management within a government organization, and a significant milestone in my journey as a developer.